

Movement and Alignment of Microtubules in Electric Fields and Electric-Dipole-Moment Estimates

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The effects of both constant (up to 2×10^3 V/m) and high-frequency alternating fields (up to 2.1×10^5 V/m, 200 kHz to 2 MHz) on suspended microtubules are investigated. At pH 6.8 and 120 mM ionic strength, constant fields cause a motion of microtubules toward the anode. The electrophoretic mobility amounts to 2.6×10^{-4} cm²/Vs, reflecting a negative net charge of approximately 0.2 elementary charges per tubulin dimer. The moving microtubules are randomly space oriented. Alternating high-frequency fields induce electric dipoles and align the microtubules parallel to the field direction. By determining the angular velocity of the turning microtubules, we estimate a dipole moment of roughly 34,000 Debye at 2.1×10^5 V/m and 2 MHz. By comparing the potential energy of the dipole in the applied field with the thermal energy of microtubules, we obtain a minimum value of 6,000 Debye as necessary for an efficient alignment.

Keywords Alignment; Electric fields; Electro-orientation; Induced dipoles; Microtubules.

Introduction

Microtubules are ubiquitous eukaryotic proteinaceous filaments, which are involved in different cellular functions including segregation of genetic material, cell morphogenesis, and intracellular vesicle transport. Microtubules consist of alpha-beta tubulin dimers, arranged in a chain-like configuration to form the so-called protofilaments (for assembly pathway see e.g., Unger et al. [1]). Due to the dimer nature of the tubulin and the strict alpha-beta alternation, each protofilament terminates in an alpha tubulin at one end, and in a beta tubulin at

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the other. This gives the protofilament a structural and functional polarity. The complete microtubular wall then is formed by a parallel isopolar arrangement of a definite number of such protofilaments (13 in the case of mammalian cytoplasmic microtubules). As a consequence of such an organization, the microtubule itself becomes polar, with the beta tubulin subunits exclusively at one end (“plus end”), and the alpha tubulin subunits at the other (“minus end”). This polarity provides the basis for direction-dependent microtubule functioning.

Under cell-free conditions, microtubules can be reconstituted by self-assembly in the presence of guanosine triphosphate (GTP) and magnesium ions at physiological temperatures. The regularly organized microtubule structures with nanometer-sized tubulin and protofilament subunits imply a potential role for microtubules as novel components to be involved in bio-nanotechnological developments [2–8]. One key problem of integrating microtubules into micro- and nano-devices is to regulate the orientation and positioning of the microtubule biopolymers on a given matrix. A first successful solution to align a great number of microtubules was described by Stracke et al. [9], who applied a hydrodynamic procedure to steer microtubules actively driven by the motorprotein kinesin across a glass surface. As a result, 2-dimensional arrays were produced with microtubules oriented in parallel and isopolar manner.

The isoelectric points of alpha and beta tubulin are found at pH 5.45–5.65 and pH 5.30–5.45, respectively [10, 11]. Interestingly, the isoelectric point for (taxol-stabilized) microtubules was found to be much lower, at pH 4.2 [12]. Both alpha and beta tubulin reveal distinct negatively charged domains, comprising about 40 residues at the COOH-terminus [13]. This negative charge is preferentially located on the surface of the microtubules. Due to this negative net charge and the postulated dipole character of tubulin [14–16, 32], suspended microtubules should interact correspondingly with electric fields under conditions of an incomplete charge neutralisation by counterions of the buffer solution.

In electrically neutral particles, electric fields induce a dipole moment by partial separation of opposite charges (for the following theoretical principles see Bakewell [17]):

$$\vec{p} = \alpha V \vec{E}$$

$\vec{p}$: vector of dipole moment

$\vec{E}$: vector of electric field (root mean square (rms)-value for alternating fields)

V : volume

α : induced polarisability $\alpha(\omega) = 3\epsilon_m \text{Re}\langle(\epsilon_p^* - \epsilon_m^*)/(\epsilon_p^* - 2\epsilon_m^*)\rangle$

The term $(\epsilon_p^* - \epsilon_m^*)/(\epsilon_p^* - 2\epsilon_m^*)$ is the so-called Clausius-Mosotti factor.

ϵ : permittivity

$\epsilon^* = \epsilon - j\sigma/\omega$ (* denotes the complex character of the quantity)

$\omega = 2\pi f$ (f : frequency)

Re: real part of the complex quantity

The indices “ p ” and “ m ” indicate “particle” and “medium”

A torque is generated by the interaction of the induced dipole with the electric field, leading to an alignment of an (elongated) particle parallel to the field lines:

$$\vec{T} = \vec{p} \times \vec{E}$$

$$|T| = pE \sin \beta$$

$\vec{T}$: vector of torque

$\vec{p}$: vector of dipole moment

E : vector of electric field

β : angle between the vectors of $\vec{p}$ and $\vec{E}$.

This so-called electro-orientation was shown for different biological objects with nonspherical shape [18, 19]. If the electric field is nonuniform, apart from inducing the torque, it also exerts a force on the particle and moves it along the direction of the field gradient. This effect is known as dielectrophoresis [19].

This article describes the effects of constant and alternating electric fields on suspended microtubules. In addition, we provide conditions for using electric fields to steer microtubule orientation generally and to obtain ordered, 2- and 3-dimensionally organized arrays. Besides the bio nano-technological aspects of this study, the results might contribute to a better understanding of the electric properties of microtubules and their behaviour in electric fields, and consequently to improve the understanding of molecular mechanisms of field-influenced microtubule-dependent processes within cells.

Materials and Methods

Preparation of Microtubules

The tubulin was purified from porcine brain by 2 cycles of temperature-dependent disassembly/reassembly [20], followed by phosphocellulose column chromatography [21].

Microtubule stock suspensions were prepared from tubulin (6.9 mg/ml) by 30-min incubation at 37°C in PEM buffer (20 mM PIPES, 80 mM NaCl, 1 mM EGTA, 0.5 mM MgCl₂, pH 6.8) supplemented with 1 mM GTP and 70 μM paclitaxel (SIGMA-ALDRICH Chemie, Taufkirchen, Germany) as assembly promoter. The paclitaxel is required also for reducing treadmilling and the dynamic instability in the microtubule suspension, and for stabilizing individual microtubules against disassembly, which might occur upon dilution or as a result of electric field-induced changes of ionic strength.

Visualization of Microtubules

For the investigations with constant fields, the microtubule stock suspensions were diluted 40 to 100-fold with the PEM buffer (ionic strength: about 120 mM), containing 10 μM paclitaxel.

For the investigations with alternating fields, the stock solutions were diluted 3–5-fold by distilled water containing 20 μM paclitaxel. Dilution was necessary to get an appropriate microtubule density, permitting the visualization of individual microtubules, and also to reduce the electric conductivity of the buffer to about 5 mS/cm.

The microtubules were visualized by an Axiophot microscope (ZEISS, Jena, Germany), equipped with a Chalnicon video camera, type C2400-01 and the image processing system Argus 20 (both obtained from Hamamatsu Photonics Deutschland, Herrsching, Germany). Image processing was performed according to Weiss and Maile [22].

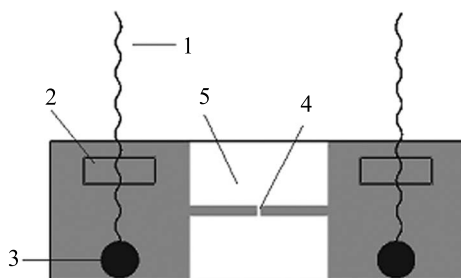


Figure 1. Electrode system for studying suspended microtubules in alternating electric fields. Grey areas: gold layer, 1: thin, flexible wires for application of the voltage, 2: strips of adhesive tape as pull relief, 3: drops of conductive silver for contacting the wires with the gold, 4: electrode gap filled with the sample, 5: gold-free glass areas.

Microtubule Suspensions in the Electric Fields

Constant Electric Fields. Microtubule suspensions were investigated in constant electric fields using glass chambers ($2.2\text{ mm} \times 50\text{ mm}$, height: $100\text{ }\mu\text{m}$) prepared on microscopic slides. Platinum electrodes were fixed in 2 plastic block containers of about 1 ml volume, functioning as buffer reservoirs, and separated by agarose gel (1% in PEM buffer) from the chamber. The voltage was varied in the range between 0 and 100 V, resulting in maximum field strength of about 2000 V/m (for details see Stracke et al. [12]).

Alternating Electric Fields. A microscopic slide ($26\text{ mm} \times 76\text{ mm}$) was used as a support to deposit the electrode system of a thin gold layer prepared by sputtering (see Fig. 1). The gap width between the electrodes was about $50\text{ }\mu\text{m}$.

To the gap region $2\text{ }\mu\text{l}$ of the microtubule suspension was applied and covered by a cover slip ($18\text{ mm} \times 18\text{ mm}$), which was sealed by wax to prevent drying of the sample. The voltage generator provided a maximum alternating voltage of 80 V pp (peak to peak) in the frequency range between 200 kHz and 2 MHz (sinusoidal voltage).

Results

Transport of Microtubules in Constant Electric Fields

Application of constant electric fields in the range of 100 V/m to 2000 V/m results in a transport of suspended microtubules to the anode due to a negative net charge of the microtubules under the used conditions. The electrophoretic mobility (B) was determined by measuring the velocity of single microtubules in the field:

$$B = v/E$$

v : migration velocity,
 E : electric field strength.

In addition to the measurement of the microtubule migration velocity, the electric current also was determined, which allows the calculation of the electric field strength:

$$E = I\rho/bh$$

I : electric current,

ρ : specific resistance,

b and h : channel width and height, respectively.

Our results yield an electrophoretic mobility of $(2.6 \pm 0.4)10^{-4} \text{ cm}^2/\text{Vs}$ at pH of 6.8, an ionic strength of 120 mM, and a temperature of 25°C. To exclude the influence of the electro-osmotic flow, the measurements had to be carried out in the so-called stationary layers defined according to Komagata [23].

A preferred microtubule orientation could not be observed at the relatively low field strength used. To realize higher values of the field strength, the same electrode system as described for generation of the alternating high frequency fields (with 50- μm distance between cathode and anode) was used at a voltage of 1 V, theoretically resulting in a field strength of $E = 2 \times 10^4 \text{ V/m}$, which is below the decomposition voltage. But, since the effect of electrode polarization fully compensates the influence of the applied voltage only at the moment following switching on or off the voltage, the microtubules briefly moved. Later, only Brownian motion, that is, without any directed motion or orientation, could be seen.

Alignment of Microtubules by Alternating Electric Fields

Alternating electric fields gradually turn suspended microtubules from an initially random orientation at zero field strength toward the field direction and align them parallel to each other. This effect starts at a field strength of approximately $1 \times 10^5 \text{ V/m}$ and becomes evident above $2 \times 10^5 \text{ V/m}$ (Fig. 2). The selected frequency ranges from 200 kHz to 2 MHz. At $2.1 \times 10^5 \text{ V/m}$ and 2 MHz, the time to turn the suspended microtubules from a rectangular orientation relative to field direction to a parallel orientation was about one second.

The alignment was observable in the whole frequency range, but at 2 MHz the alignment appeared to be somewhat better compared to lower frequencies. Typical microscope images for different field conditions are given in Fig. 2.

Estimation of the Induced Dipole Moment

Estimation of the angular velocity of single microtubules during alignment enables one to get the induced dipole moment to a rough approximation. The free rotation of a thin cylindrical rod in a viscous fluid with constant angular velocity needs a torque T_{visc} of:

$$T_{\text{visc}} = \gamma\eta\Omega L^3$$

$\gamma = (\pi/3)/\ln(L/2r) - 0.66$ (constant shape factor) (Howard, 2001)

L , r : length and radius of the cylinder, respectively

Ω : angular velocity

η : dynamic viscosity.

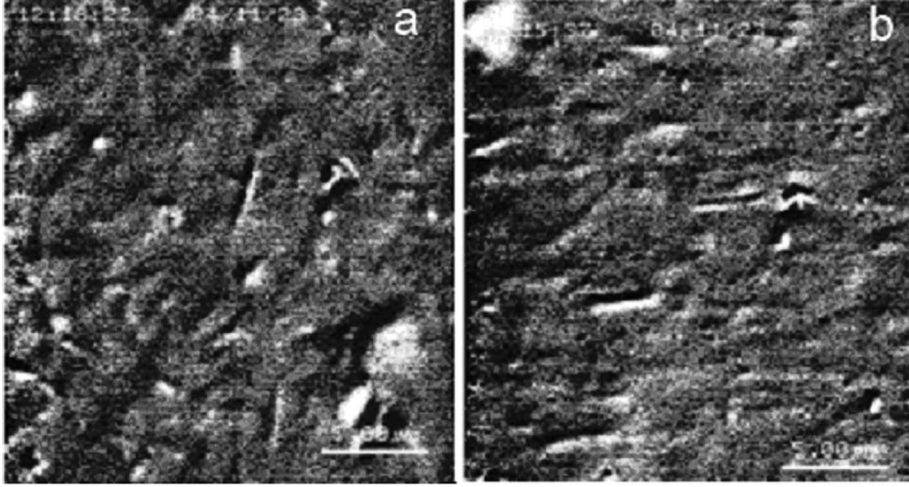


Figure 2. Alignment of suspended microtubules by high frequency electric fields. a) Control (without field), that state is realized again after switching off the alternating field; b) $E_{\text{rms}} = 210000 \text{ V/m}$, 2 MHz. Microtubule visualization by video-enhanced DIC microscopy.

The torque $\vec{T}_{\text{el}}$ exerted on a dipole in an electric field is:

$$\vec{T}_{\text{el}} = \vec{p} \times \vec{E}$$

$$|T| = pE \sin \beta$$

$\vec{p}$: vector of dipole moment

E : vector of electric field strength

β : angle between $\vec{p}$ and $\vec{E}$.

In the case of an alternating field with sinusoidal time dependence, E has to be replaced by the root mean square (rms)-value (E_{rms}).

As the torque depends on the angle between the dipole and the field vector, a mean value of the torque (in the angular range between 0 and $\pi/2$) is calculated:

$$T_{\text{mean}} = 0.637 p E_{\text{rms}}$$

In the stationary state the torques T_{visc} and $\vec{T}_{\text{el}}$ must be equal, which yields

$$\gamma \eta \Omega L^3 = 0.637 p E_{\text{rms}} \quad \text{or} \quad p = \gamma \eta \Omega L^3 / 0.637 E_{\text{rms}}.$$

The dipole moment in the axial direction of the cylinder (i.e., the microtubule) cannot be a constant quantity, as the alignment starts from a state in which the axial dipole moment is nearly zero (i.e., cylinder and field are perpendicular to each other).

Therefore, only a mean value, p_{mean} , of the dipole moment can be calculated. For the same reason the value of Ω , estimated from video records, has to be regarded as a mean value, Ω_{mean}

$$p_{\text{mean}} = \gamma \eta \Omega_{\text{mean}} L^3 / 0.637 E_{\text{rms}}$$

The time required to turn microtubules so as to achieve an alignment starting from a position approximately perpendicular to the field was found to be about one second.

For microtubules with a length of $L = 3.5 \cdot 10^{-6}$ m (mean length), $\Omega_{\text{mean}} \approx \pi/2 \text{ s}^{-1}$, $\gamma = 0.245$, $\eta = 0.925 \cdot 10^{-3}$ kg/ms and $E_{\text{rms}} = 2.12 \cdot 10^5$ V/m an induced electric dipole moment

$$p_{\text{mean}} \approx 1.1 \cdot 10^{-25} \text{ Asm} \approx 34,000 \text{ Debye}$$

results, formally corresponding to about 0.2 elementary charges at the ends of the microtubules.

It must be considered that the thermal movement of the suspended microtubules counteracts their orientation in the field. To evaluate the relationship between the thermal movement and the dipole-field interaction we compare the potential energy of a dipole in the field with the thermal energy of the dipole:

$$E_{\text{pot}} = pE(1 - \cos \beta)$$

β : angle between the directions of the vectors $\vec{p}$ and $\vec{E}$ ($0 \leq \beta \leq \pi$).

At a mean value of $\beta = \pi/2$

$$E_{\text{pot}} = pE.$$

If for the rotational energy two degrees of freedom (each with $KT/2$) are assumed, the thermal energy has the value of

$$E_{\text{therm}} = kT$$

$k = 1.38 \cdot 10^{-23}$ Js/K (Boltzmann's constant)

T : Temperature

For $E_{\text{pot}} = E_{\text{therm}}$ it then follows that $p \approx kT/E$.

At room temperature $T = 298$ K and an applied electrical field of $E = 2 \times 10^5$ V/m the lower limit of the dipole moment amounts to

$$p_{\text{min}} \approx 6000 \text{ Debye.}$$

This means that an alignment of microtubules can be expected at dipole moments $> 6,000$ Debye, only. Under our experimental conditions, an effective alignment was achieved at 34,000 Debye.

Discussion

The main effects of electric fields on particles such as microtubules are transport and orientation. If the particles carry an excess electric charge they will be forced to move in an electric field by the force

$$F = qE$$

q : electric charge,

E : electric field strength.

Due to the retarding effects of viscous fluids on moving particles their velocity is constant and can be characterized by the electrophoretic mobility

$$B = v/E$$

v : migration velocity.

We studied the effects of both constant electric fields (up to 2×10^3 V/m) and high-frequency alternating electric fields (up to 2.1×10^5 V/m with frequencies between 200 kHz and 2 MHz) on suspended microtubules.

Application of constant voltages below the decomposition voltage of approximately 1 V at a small electrode distance (50 μ m), aiming to enhance the field strength without electrolysis, was not successful. The field strength is reduced to nearly zero by electrode polarization. Consequently, we could not observe a microtubule transport under such conditions with the exception of a very short time interval after the switching on or off of the voltage. In contrast to our results, van den Heuvel et al. [24] and Jia et al. [25] found a transport of microtubules in constant electric fields below the decomposition voltage.

A neutral particle with separated positive and negative charges has a dipole moment of

$$p = qL$$

q : charge,

L : distance between both charges.

Such a dipole cannot be moved in a uniform electric field. The sum of the forces exerted on the positive and the negative charge is zero. But, the interaction of an electric dipole with an electric field results in a torque

$$\vec{T} = \vec{p} \times \vec{E}$$

$\vec{T}$, $\vec{p}$, $\vec{E}$: vectors of torque, dipole moment, electric field, respectively which causes the dipole to be aligned parallel to the electric field.

Brown [26] calculated the permanent dipole moment of the tubulin dimer from molecular data. Merzhin et al. [27] took into account the screening effect of water (whose dielectric constant may be as high as 80) on the dimer charges, which yields the estimate for the (permanent) dimer dipole moment

$$p_{\text{dimer}} = 3 \times 10^{-28} \text{ C m} = 90 \text{ Debye}$$

although, “unpaired charges,” that is, net charges inside the hydrophobic pockets of the dimers, that appear isolated from their environment, may lead to even further suppression (note that a simplistic calculation based on unpaired charges separated by about 4 nm yields $p_{\text{dimer}} = 1,200$ Debye, while a more sophisticated molecular simulation [26, 27] could yield $p_{\text{dimer}} = 1,700$ Debye). At this point it should be stressed that, according to the analysis of Brown [26], the dimer dipole-moment vector primarily may be oriented in an axis perpendicular to the long axis of the microtubule. Indeed, theoretical estimates of the permanent dipole moment of the

dimers, taking into account their detailed structure outlined in Nogales et al. [28], yield the following values for the dipole moment $\vec{p}_{\text{dimer}}$ of the tubulin dimers:

$$p_x = 337 \text{ Debye}, \quad p_y = -1669 \text{ Debye}, \quad p_z = 198 \text{ Debye}.$$

Thus, we see that only about one-fifth of the total dipole moment of order 1,700 Debye is along the protofilament axis, which is along the x -direction. As we have discussed above, however, screening effects of the ordered water and other environments may significantly reduce the total dipole moment for the dimer to about 90 Debye. From the estimates of Brown [26] above, one-fifth, or realistically less, of that may be directed along the microtubule axis, that is, one may take as an order of magnitude 15 Debye.

One may speculate that the other components perpendicular to the microtubule axis will be neutralized in the cylindrical microtubular geometry and screened by the environment, thereby leaving the p_x component as the dominant contribution to the total dipole moment of the microtubule. If this is the case, then taking into account that in moderately long microtubules of an average length $L = 3.5 \mu\text{m}$, with 12 protofilaments each (as we used in our experiments), there are about $N = 3.5 \mu\text{m} / 8 \text{ nm} \times 12 = 5280$ tubulin dimers of average length 8 nm, one would arrive at the most optimistic estimate for the total dipole moment (all dimer dipoles contributing equally to the x -direction) in the region

$$p_{\text{total}} = Np_{\text{dimer}} = 79,200 \text{ Debye},$$

which is likely to be further suppressed if details on the water and other environments and geometry are properly taken into account.

Such a suppression of the total dipole moment may provide an explanation for our inability to observe an alignment of microtubules in constant electric fields up to $2 \times 10^3 \text{ V/m}$. Estimates of the interaction energy of the supposed permanent dipoles in the electric fields show that this energy is too small to overcome the influence of the thermal energy [32].

The question of whether or not constant fields of higher strength (e.g., $2 \times 10^5 \text{ V/m}$, as in the successful case of alternating fields, see above) would be capable of aligning the microtubules could not be answered within the frame of this study due to safety requirements. However, it was demonstrated that even relatively weak constant electric fields (1250 V/m) are able to steer microtubules driving across a suitable substrate by the motor protein kinesin [12]. Due to kinesin-mediated binding of the microtubules to the substrate Brownian movement is strongly reduced. In the absence of the field, microtubule gliding, which is restricted to a 2-dimensional layer, does not have a preferential direction. When the field is applied, up to 80% of the microtubules are forced to move to the anode, resulting in a roughly parallel, isopolar alignment. A complete alignment was not achieved due to the fact that a force arising from the electro-osmotic flow near the substrate surface counteracts the steering effect of the field.

The use of high-frequency alternating fields compared to constant fields has the advantage that there are no problems with electrolysis and hence high electric field strengths can be applied. Under the influence of alternating electric fields, induced dipoles are oriented and also forced to move, depending on the field nature. Indeed, uniform alternating fields exert orientating torques only, whereas

inhomogeneous alternating fields move the dipole-carrying particle along the direction of the field gradient (or in the opposite direction, depending on the sign of the induced polarization, see below). Most importantly, this force is independent of the field direction (note that both field and dipole moment change their signs simultaneously).

We found, that alternating high frequency fields (2×10^5 V/m, 2 MHz) align microtubules and turn them parallel to the exerted field. By determining the angular velocity of turning microtubules during alignment we estimated an induced dipole moment of 34,000 Debye, corresponding to an effective charge separation of approximately 0.2 elementary charges. This value depends on both physical and chemical parameters such as electric field strength, frequency, pH, and ionic strength.

Independent of the fact that our investigations provide no experimental evidence for the permanent dipole, a possible permanent part of the dipole moment does not play a role for the orientation at high frequencies because the dipole cannot follow the changes of the field. Only the induced part of the dipole is responsible for the orientation because the torque does not depend on the field direction. Comparing the potential energy of the dipole in the applied field with the thermal energy of microtubules shows that a minimum value of the dipole moment is necessary for a successful orientation.

Conclusions

In this work it was demonstrated that suspended microtubules are affected by both constant and alternating electric fields. Contrary to the constant fields that move microtubules without a preferred orientation to the anode, application of the alternating fields leads to parallel orientation of microtubules due to induced dipole moments. By measuring the angular velocity of the turning microtubules during alignment, we estimated these dipole moments to be of order 34,000 Debye. It has to be mentioned that this value is valid only for the exact conditions used in this study. Further experiments with other physical and chemical parameters are necessary to get more general results. To measure the permanent dipole moment other methods have to be used [33, 34].

The alignment of microtubules in alternating high-frequency fields provides a methodological basis to obtain 3-dimensional arrays of microtubules with geometrically parallel orientation.

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